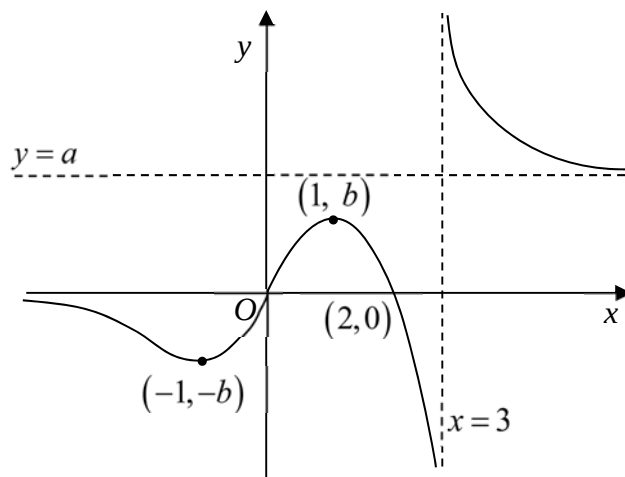


- 1** One of the roots of the equation $w^2 - (6 - 5i)w + 11 - i = 0$ is $1 + i$.
 Explain why the Conjugate Root Theorem cannot be applied to find the other root. [1]
- Determine the other root of the equation $w^2 - (6 - 5i)w + 11 - i = 0$ in the form $p + iq$. [2]
- 2** (i) State a sequence of transformations which transforms the graph of $y = x^2$ to the graph of $y = \frac{1}{a}(x - a)^2 + \frac{1}{a}$, where a is a positive constant. [3]
- (ii) A curve has equation $y = f(x)$, where
- $$f(x) = \begin{cases} x^2 & \text{for } -1 < x \leq 1, \\ \frac{1}{2}(x - 2)^2 + \frac{1}{2} & \text{for } 1 < x \leq 3, \\ 1 & \text{otherwise.} \end{cases}$$
- Sketch the curve of $y = f(x)$ for $-2 \leq x \leq 5$.
 You should state clearly the coordinates of the stationary points. [3]
- 3** The complex number z is given by $z = 1 + ia$, where a is a positive real number.
- (i) Given that the argument of $(z - 3)$ is $\frac{3\pi}{4}$, find the value of a . [2]
- The complex number w is given by $w = ib$, where b is a positive real number. Find
- (ii) the modulus of $\frac{w^n}{w^*}$ in terms of b and n , [1]
- (iii) the two smallest positive whole number values of n for which $\frac{w^n}{w^*}$ is purely imaginary. [4]

4



The diagram shows the curve $y = f(x)$. The curve has turning points at $(1, b)$ and $(-1, -b)$, and crosses the x -axis at the origin and $(2, 0)$. The curve has asymptotes $y = 0$, $y = a$ and $x = 3$. It is given that a and b are positive constants with $a > b$.

On separate diagrams, draw sketches of the graphs of

- (a) $y = \frac{1}{f(x)}$, [3]
 (b) $y = f'(x)$, [3]
 (c) $y = |f(x)|$, [2]

stating, if it is possible to do so, the coordinates of the points where the graphs cross the axes, the coordinates of any turning point(s) and the equations of any asymptotes.

5 It is given that $y^3 = 7 + \cos x$.

- (i) Show that $6y \left(\frac{dy}{dx} \right)^2 + 3y^2 \frac{d^2y}{dx^2} = 7 - y^3$. [2]
 (ii) By differentiating this result, show that the Maclaurin series for y does not have a term in x^3 . [3]
 (iii) Hence write down the series expansion of y up to and including the term in x^2 . Give the coefficients in exact form. [1]
 (iv) Using appropriate expansions from the List of Formulae (MF26), verify the expansion found in part (iii). [3]

- 6 The sequence $\{u_n\}$ is such that $u_n = \frac{ne^{n^2}}{n+1}$ where $n \in \mathbb{C}^+$. It is given that

$$\sum_{r=1}^n r^2 = \frac{n}{6}(n+1)(2n+1).$$

- (i) Find $\sum_{r=1}^n \ln(u_r)$ in terms of n . [3]

A sequence $\{v_n\}$ is such that $v_n = 2^{\ln(u_n)-n^2}$, $n \in \mathbb{C}^+$. Another sequence $\{w_n\}$ is defined as $w_n = v_1 \times v_2 \times v_3 \times \dots \times v_n$, $n \in \mathbb{C}^+$.

- (ii) Hence, or otherwise, find w_n in terms of n and determine, with a reason, if $\{w_n\}$ is convergent. [5]

- 7 (a) (i) Find $\int \cos 3x \cos x \, dx$. [3]

- (ii) Hence, find the exact value of $\int_0^{\frac{\pi}{4}} x \cos 3x \cos x \, dx$. [4]

- (b) Find, in terms of a , $\int_{-a}^a |x(x-2a)| \, dx$, where a is a positive constant. [3]

- 8 Eden is a new engineer for the Small Rail Transit (SRT) train. His current project is to devise a new schedule for the train to increase its efficiency.

In the current schedule, the train will run for 5 days in one cycle. For the first day of the first cycle, the train runs for 100 km. On the 2nd and on the 3rd day of each cycle, the train will run 30 km more than the first day of that cycle. On the 4th and on the 5th day of each cycle, the train will run 70 km more compared to the first day of that cycle. On the first day of the k^{th} cycle, the train will run 30 km more compared to the last day of the $(k-1)^{\text{th}}$ cycle.

- (i) It was found out that the train broke down after travelling 10 000 km. By forming an inequality, state the cycle and the day of the cycle that the train was in during the breakdown. [6]

Eden decides to modify the schedule such that the train will now operate every week from Monday to Saturday. On the first day of operations, a Monday, the train runs for 100 km. On every Tuesday, Thursday and Saturday, the distance travelled will be increased by 20% compared to the previous operating day. However, on every Monday, Wednesday and Friday, there will be a 8% decrease in the distance travelled compared to the previous operating day.

- (ii) Find the total distance travelled by the train in $2n$ days of operations. [3]
 (iii) Find the minimum number of days that the train has run, such that Eden would have improved the previous schedule (i.e. the total distance run before breakdown is more than 10 000 km). [2]

- 9 The two lines l_1 and l_2 have equations $x = \frac{z}{-2} - 2, y = 0$ and $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ q \\ 2 \end{pmatrix}$

respectively, where $\mu \in \mathbb{R}$ and q is a constant. The point A has coordinates $(4, 3, k)$ where k is a constant and the two planes p_1 and p_2 have equations $4x = -8 + y - 2z$ and $y + tz = 2t$ respectively, where t is a constant.

- (i) Find a vector equation of l_1 and hence show that l_1 lies in p_1 . [3]
- (ii) Find the coordinates of the foot of perpendicular from A to p_1 . Express your answer in terms of k . [3]
- (iii) Given that the angle between l_1 and l_2 is 60° , find the possible values of q . [2]
- (iv) Given that $q = 1$ and that the point $B(1, 2, 2)$ is equidistant from l_2 and p_2 , find the possible values of t . [6]

- 10 (a) A skier slides freely from rest at a particular slope at a ski resort and t seconds later, he is sliding down at a speed of $v \text{ ms}^{-1}$. He experienced gravitational force, friction and air resistance which affect v . The rate of change of v is modelled as $\frac{dv}{dt} = 7.2 - kv^2$. It is given that when his speed reaches 30 ms^{-1} , the rate of change of his speed is 5.4 ms^{-2} .

By solving the differential equation, show that $v = \frac{60(e^{mt} - 1)}{e^{mt} + 1}$, where m is a positive constant to be found. [6]

- (b) The owner of the ski resort bought new sets of ski boards. He proposes that the velocity ($y \text{ ms}^{-1}$) of a skier on the slope now satisfies this differential equation where x is the distance travelled from a starting point (x metres).

$$3x \frac{dy}{dx} = 3y - 4 + x, \text{ where } x > 1$$

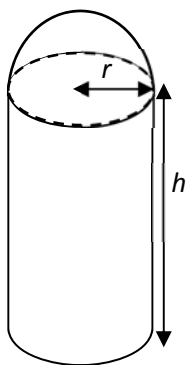
Using the substitution $y = ux$, show that the differential equation can be reduced

to $3x^2 \frac{du}{dx} = x - 4$ and hence find the general solution for y in terms of x . [5]

Suggest a suitable reason why the differential equation would not be appropriate in modelling how the velocity varies for large distances. [1]

- 11 A company designs a candy container in the shape shown below. The container is made up of an open cylinder of height h cm and radius r cm, with a hollow hemispherical lid of radius r cm. Its lid costs a cents per cm^2 , where a is a fixed constant and the cylinder costs half as much per cm^2 to manufacture. The company wants to design the container such that its total volume is 300 cm^3 . Find, using differentiation, the exact value of r and h that will minimise the cost. [7]

[Volume of sphere, $V = \frac{4}{3}\pi r^3$; Surface area of sphere, $S = 4\pi r^2$]



The annual profit made by the company is P (in ten thousand dollars per year). The economist believes that the relationship between P and t (in years) is given by

$$P = \frac{\sqrt{\ln(t+1)}}{t+1}, \text{ for } 0 \leq t \leq 5.$$

Sketch the graph of P against t , stating the coordinates of the turning point and the end points. [2]

Find, by integration, the exact area of the region bounded by the curve, the t -axis and the line $t = 5$. Give an interpretation of the area that you have found, in the context of the question. [3]

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